

Quantitative Risk Assessment of Market Volatility Under Macroeconomic Impacts

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Quantitative Risk Management Project

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- Financial markets are increasingly driven by **macroeconomic shocks** (e.g., CPI releases, FOMC rate decisions).
- Traditional risk models often assume **Normally distributed returns**.
- This assumption leads to systematic **underestimation of tail risk** during periods of market stress.

What is Tail Risk?

Definition

Tail risk is the risk of **extreme losses** that occur in the far ends (“tails”) of the return distribution, where rare events happen more often than a Normal model predicts.

- Under a Normal assumption, extreme moves are **too unlikely** $\Rightarrow$ risk is understated.
- In real markets, returns often show **fat tails** and **volatility clustering**.
- Tail risk is central to **VaR/ES**, margin, hedging, and drawdown control.

Key Macroeconomic Events: CPI and FOMC

CPI (Consumer Price Index)

A major measure of **inflation**. CPI surprises can shift expectations about interest rates and future cash flows, leading to large moves in equities and bonds.

FOMC (Federal Open Market Committee)

The Fed committee that sets **monetary policy** (e.g., target rate decisions, guidance, balance sheet policy). FOMC announcements can trigger sudden repricing of risk across markets.

- CPI → inflation expectations → rate path expectations
- FOMC → discount rates, liquidity conditions, and “risk-on/risk-off” sentiment

Project Objectives

- ① Automate market data collection and macroeconomic event mapping.
- ② Validate key statistical properties:
 - Stationarity
 - Normality
- ③ Model volatility using **GARCH(1,1)** with a Student- t distribution.
- ④ Compute and backtest **Value-at-Risk (VaR)**.

- Retrieved adjusted close prices (2022–2024) using `yfinance`.
- Collected macroeconomic events (e.g., CPI and FOMC decisions).
- Non-trading day events mapped to the **nearest subsequent trading day** to capture immediate market impact.

Stationarity

- Log-returns validated using the **Augmented Dickey-Fuller (ADF)** test.

Normality

- Shapiro–Wilk and Jarque–Bera tests applied.
- Evidence of **skewness and excess kurtosis (fat tails)**.

Why Stationarity and Normality Matter

Stationarity (core requirement for many time-series tools)

A series is **stationary** if its statistical properties (mean/variance/autocorrelation) are stable over time.

- **Why it matters:** ARIMA/GARCH inference and forecasting are more reliable when inputs are stationary.
- **Practical approach:** prices are often non-stationary $\Rightarrow$ use **log-returns** to stabilize behavior.

Normality (often assumed, often violated)

Normality implies thin tails and symmetric returns, which can underestimate extremes.

- **Why it matters:** VaR under Normality can be **too small** during stress periods.
- **Response:** use Student- t innovations to better capture fat tails.

Skewness and Excess Kurtosis (Fat Tails)

Skewness (asymmetry)

- **Left-skew (negative skew):** more probability mass in the **left tail**
⇒ larger downside moves.
- **Right-skew (positive skew):** more probability mass in the **right tail**
⇒ occasional large gains.

Excess Kurtosis (tail heaviness)

Excess kurtosis > 0 indicates **fat tails**: extreme outcomes occur more often than in a Normal distribution.

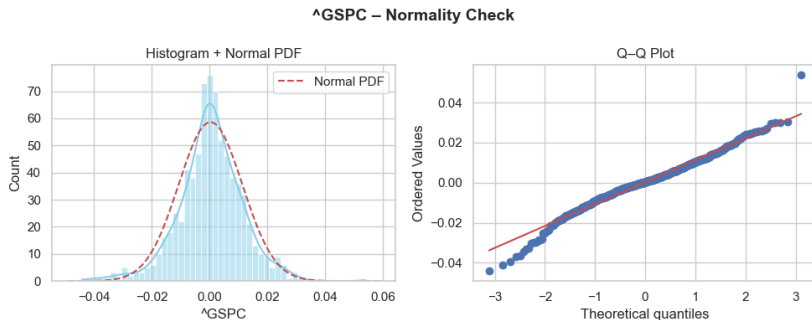
- Fat tails ⇒ more VaR breaches under Normal assumptions
- Student- t innovations help model tails more realistically

- **ARIMA(1,1,0):**
 - Modeled the linear trend of log-prices.
- **GARCH(1,1)** with Student- t innovations:
 - Modeled conditional volatility.
 - Captured volatility clustering and fat tails.

Risk Metrics and Backtesting

- Computed **95% Value-at-Risk (VaR)** using dynamic volatility forecasts.
- Backtested VaR with the **Kupiec Proportion of Failures Test**.

Figure 1: S&P 500 Normality Check

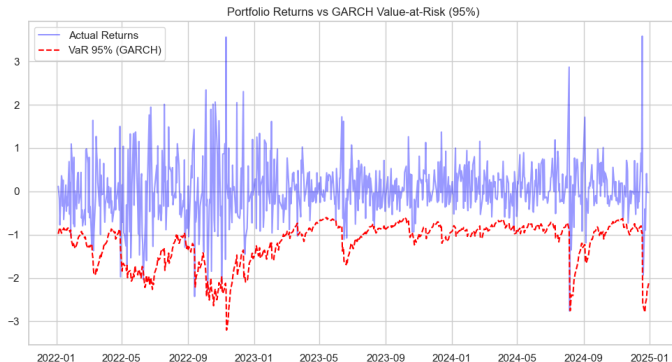


The Histogram (left) shows a peak higher than the Normal PDF. The Q-Q Plot (right) shows data points diverging from the red line at the extremes, confirming fat tails.

Results: Key Findings

- **Non-Normality confirmed:** heavy tails justify Student- t innovations.
- **Dynamic risk:** GARCH adapts to volatility clusters.
- During stress periods (e.g., 2022 rate hikes), VaR expands to widen the safety buffer.

Figure 2: Portfolio Returns vs. 95% GARCH VaR



The red dashed line (VaR) dynamically adjusts to market volatility. Red dots indicate days where actual losses exceeded the model's prediction.

- Kupiec backtest yields **p-value** > 0.05 .
- We **fail to reject** the model (statistically valid for this period).

Implications: What Would You Do During Volatile Macro Events?

Model-driven insight

If volatility is time-varying and tails are fat, risk controls should be **dynamic**, not static.

- **Risk budgeting:** reduce exposure when forecast volatility rises (de-risk / tighten limits).
- **Hedging:** increase protective hedges around high-impact releases (e.g., CPI, FOMC), especially when model-implied tail risk is elevated.
- **Position sizing:** scale position sizes using volatility forecasts (e.g., volatility targeting).
- **Liquidity and leverage:** lower leverage and hold more liquidity buffers during event windows.
- **Stress testing:** supplement VaR with scenario shocks (rate surprise, equity gap-down).

- **Asymmetric volatility models:** EGARCH, GJR-GARCH (leverage effect).
- **Expected Shortfall (ES):** complements VaR and better captures tail losses.

Conclusion

- Built a reproducible workflow for quantitative risk management.
- Dynamic volatility modeling improves downside risk estimation vs. static variance assumptions.
- Accounting for **fat tails** is essential for accurate modern risk estimation.

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